10 (a) The magnitude of the charge on a β (beta)-particle is 1.6×10^{-19} C.

(i) State the proton number and nucleon number of an α (alpha)-particle.

proton number .....

nucleon number .....
[2]

(ii) Determine the magnitude of the charge of an α (alpha)-particle.

charge .....
[1]

(b) A nucleus of radium-230 consists of 88 protons and 142 neutrons. Radium-230 is radioactive and decays by β (beta)-emission to an isotope of actinium. The symbol for radium is Ra and the symbol for actinium is Ac.

Write down the nuclide equation for this decay.

[3]

(c) The half-life of radium-230 is 93 min. A sample contains 9.6×10^{-12} g of radium-230.

Calculate the mass of radium in the sample after 279 min.

mass = ... [2]

[Total: 8]

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